

Manufacturing Operations Data

Operation	Material	Machine	Duty	Setup Time (min)	Cycle Time (sec)	Work Hours	Diameter (mm)	Length (mm)	Breadth (mm)	Height (mm)
Drilling	Steel	Drilling machine	Heavy duty	25	120	1800	22	65	-	-
Turning	Aluminium	CNC	Medium duty	30	90	2400	40	120	-	-
Milling	Steel	3 axis CNC	Heavy duty	45	150	3600	-	200	60	50
Grinding	Steel	Special Purpose Machine(SPM)	Heavy duty	35	180	4000	25	80	-	-
Jig boring	Titanium	CNC	Heavy duty	50	210	3200	18	90	-	-
Rubber press	Steel	Small (less than 650 MMX 650 MM)	Heavy duty	40	300	2800	150	400	-	-
Rubber press	Aluminium	Medium (more than 650 MMX 650 MM)	Heavy duty	55	280	3500	220	600	-	-
Drilling	Aluminium	Drilling machine	Heavy duty	15	60	1200	12	50	-	-
Turning	Steel	CNC	Heavy duty	28	110	2600	55	180	-	-
Milling	Titanium	5 axis CNC	Heavy duty	60	240	4200	-	300	70	65
Grinding	Steel	Special Purpose Machine(SPM)	Heavy duty	32	160	2900	30	100	-	-
Jig boring	Steel	CNC	Heavy duty	48	200	3300	20	85	-	-
Rubber press	Steel	Medium (more than 650 MMX 650 MM)	Heavy duty	58	320	4100	250	750	-	-

Operation	Material	Machine	Duty	Setup Time (min)	Cycle Time (sec)	Work Hours	Diameter (mm)	Length (mm)	Breadth (mm)	Height (mm)
Drilling	Steel	Drilling machine	Heavy duty	22	95	1900	18	60	-	-
Turning	Aluminium	Conventional	Light duty	18	70	1500	35	140	-	-
Milling	Steel	CNC	Medium duty	40	130	2700	-	160	45	40
Grinding	Titanium	Special Purpose Machine(SPM)	Heavy duty	38	190	3600	28	95	-	-
Jig boring	Steel	CNC	Heavy duty	52	230	3900	24	110	-	-
Rubber press	Steel	Small (less than 650 MMX 650 MM)	Heavy duty	42	260	3000	180	500	-	-
Rubber press	Aluminium	Medium (more than 650 MMX 650 MM)	Heavy duty	60	340	4500	300	900	-	-